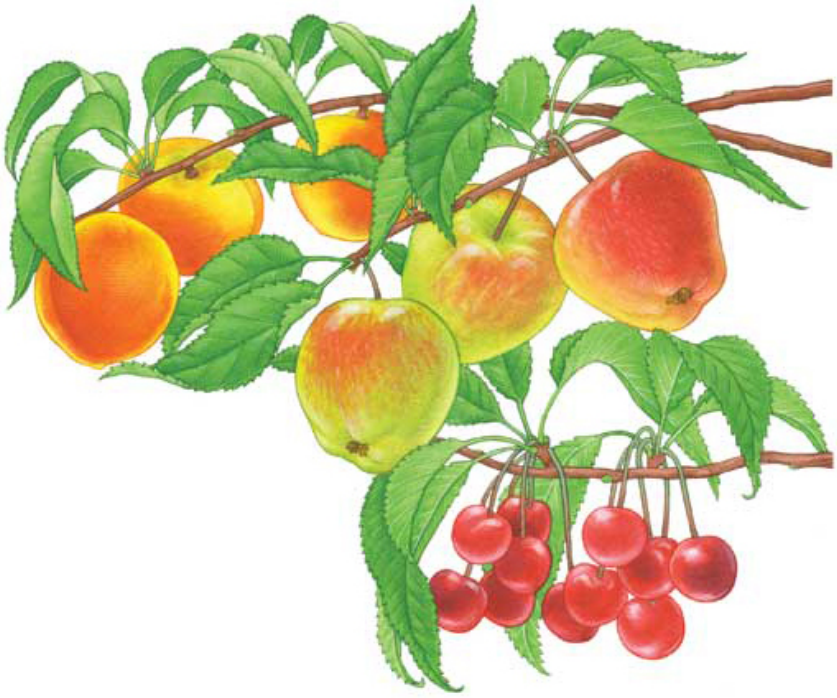
MOTHER EARTH NEWS

All About Growing Fruit Trees

Growing fruit trees organically is possible with the proper amount of care and attention. To bite into a fresh peach, or spread homemade apple butter on warm bread, is the epitome of a sweet, sweet reward.
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There are many types of fruit trees, and with a little research you can easily find the best variety for your region and tastes. Try growing apples for homemade cider or growing peaches for a heavenly summer treat.

Illustration by Keith Ward

(For details on growing many other vegetables and fruits, visit our Crop at a Glance collection page.)

No plants give sweeter returns than fruit trees. From cold-hardy apples and cherries to semi-tropical citrus fruits, fruit trees grow in nearly every climate. Growing fruit

trees requires a commitment to pruning and close monitoring of pests, and you must begin with a type of fruit tree known to grow well in your area.

Choose varieties recommended by your local extension service, as some varieties need a certain level of chill hours (number of hours below 45 degrees Fahrenheit). For complete details on planning and maintaining a home orchard, we recommend the book The Holistic Orchard by Michael Phillips.

Types of Fruit Trees to Try

Even fruit trees described as self-fertile will set fruit better if grown near another variety known to be a compatible pollinator. Extension publications and nursery catalogs often include tables listing compatible varieties.

Apples (*Malus domestica*) are the most popular tree fruits because they are widely adapted, relatively easy to grow and routine palate-pleasers. The ideal soil pH for apples is 6.5, but apple trees can adjust to more acidic soil if it's fertile and well-drained. Most apple varieties, including disease-resistant 'Freedom' and 'Liberty,' are adapted to cold-hardiness Zones 4 to 7 (if you don't know your Zone, see "Know Your Cold-Hardiness Zone" later in this article), but you will need low-chill varieties, such as 'Anna' and 'Pink Lady,' in mild winter climates. No matter your climate, begin by choosing two trees that are compatible pollinators to get good fruit set. Mid- and late-season apples usually have better flavor and store longer compared with early-season varieties.

Cherries (*Prunus avium* (sweet) and *P. cerasus* (sour)) range in color from sunny yellow to nearly black and are classified in two subtypes: compact sweet varieties, such as 'Stella,' and sour or pie cherries, such as 'Montmorency' and 'North Star.' Best adapted to Zones 4 to 7, cherry trees need fertile, near-neutral soil and excellent air circulation. Growing 12-foot-tall dwarf cherry trees of either subtype will simplify protecting your crop from diseases and birds, because the small trees can be covered with protective netting or easily sprayed with sulfur or kaolin clay.

Citrus fruits (*Citrus* hybrids), including kumquat, Mandarin orange, satsuma and 'Meyer' lemon, are among the easiest fruit trees to grow organically in Zones 8b to 10. Fragrant oils in citrus leaves and rinds provide protection from pests, but cold tolerance is limited. 'Nagami' kumquat, 'Owari' satsuma and 'Meyer' lemon trees may occasionally need to be covered with blankets when temperatures drop below freezing, but winter harvests of homegrown citrus fruits will be worth the trouble.

Peaches and nectarines (*Prunus persica*) are on everyone's want list, but growing these fruit trees organically requires an excellent site, preventive pest management and some luck. More than other fruit trees, peach and nectarine trees need deep soil with no compacted subsoil or hardpan. Peaches and nectarines are best adapted to Zones 5 to 8, but specialized varieties can be grown in colder or warmer climates. Peach and nectarine trees are often short-lived because of wood-boring insects, so plan to plant new trees every 10 years.

Plums (*Prunus* species and hybrids) tend to produce fruit erratically because the trees often lose their crop to late freezes or disease. In good years, plum trees will yield heavy crops of juicy fruits, that vary in color from light green to dark purple. Best adapted to Zones 4 to 8, plum trees need at least one compatible variety nearby to ensure good pollination. In some areas, selected native species, such as beach plums in the Northeast or sand plums in the Midwest, may make the best homestead plums.

Pears (*Pyrus* species and hybrids) are slightly less cold-hardy than apples but are easier to grow organically in a wide range of climates. In Zones 4 to 7, choose pear varieties that have good resistance to fire blight, such as 'Honeysweet' or 'Moonglow.' In Zones 5 to 8, Asian pear trees often produce beautiful, crisp-fleshed fruits if given routine care. Most table-quality pears should be harvested before they are fully ripe.

How to Plant

The best time to plant fruit trees in Zones 3 through 7 is early spring, after the soil has thawed. Fruit trees that are set out just as they emerge from winter dormancy will rapidly grow new roots. In Zones 8 to 10, plant new trees in February. Choose a sunny site with fertile, well-drained soil that's not in a low frost pocket. Dig a planting hole that's twice the size of the root ball of the tree. Carefully spread the roots in the hole, and backfill with soil. Set trees at the same depth at which they grew at the nursery, taking care not to bury any graft union (swollen area) that's on the main trunk. Water well, and install a trunk guard made of hardware cloth or spiral plastic over the lowest section of the trunk to protect it from insects, rodents, sunscald and physical injuries. Stake the tree loosely to hold it steady. Mulch over the root zone of the planted trees with wood chips, sawdust or another slow-rotting mulch. Water particularly well during any dry spells for the first two years.

One year after planting, fertilize fruit trees in spring by raking back the mulch and

scratching a balanced organic fertilizer into the soil surface (follow application rates on the product's label). Then add a wood-based mulch to bring the mulch depth up to 4 inches in a 4-foot circle around the tree. After two years, stop using trunk guards and instead switch to coating the trunks with white latex paint to defend against winter injuries. Add sand to the paint to deter rabbits and voles.

Pruning Fruit Trees

Pruning is a key aspect of growing fruit trees. The goal of pruning fruit trees is to provide the leaves and fruit access to light and fresh air. The ideal branching pattern varies with species, and some apple and pear trees can be pruned and trained into fence- or wall-hugging espaliers to save space. Begin pruning fruit trees to shape them in their first year, and then prune annually in late winter, before the buds swell. Pruning a little too much questionable growth is better than removing too little.

Many fruit trees set too much fruit, and the excess should be thinned. Asian pear trees should have 70 percent of their green fruits snipped off when the pears are the size of a dime, and apples should be thinned to 6 inches apart before the fruits are the size of a quarter. When any type of fruit tree is holding a heavy crop, thinning some of the green fruits will increase fruit size, reduce limb breakage and help prevent alternative bearing (a tree setting a crop only every other year).

Harvesting and Storage

With the exception of pears, tree fruits should be harvested just as they approach full ripeness and then kept chilled to slow spoilage. The flavor of most apples improves after a few weeks in cold storage, so a second refrigerator or a root cellar may be useful. Apples and pears can be kept for several months in a refrigerator, but softer stone fruits (cherries, nectarines, peaches and plums) must be canned, dried, frozen or juiced within a few days of harvest for long-term storage.

Pest and Disease Tips

Some types of fruit crops attract a large number of insect pests and can succumb to several widespread diseases for which no resistant varieties are available. For example, all of the stone fruits are frequently affected by brown rot, a fungal disease that overwinters in mummified fruit. Apply early-season sulfur sprays to suppress

brown rot and other common diseases. Some apples have good genetic resistance to scab and rust, but you will still need to manage insect pests, such as codling moths.

Allowing chickens to forage beneath fruit trees can help suppress insects. Many organic growers also keep their fruit trees coated with kaolin clay during the growing season to repel pests.

In the Kitchen

Managing the hefty harvests from mature fruit trees will require a range of food-preservation skills. Manual fruit-processing equipment, such as a cherry pitter (see Slideshow) or an apple peeler/corer, can be excellent investments. At each picking, a number of bruised fruits will need attention right away, while you can refrigerate and forget about sound fruits until your next preservation project. In addition to making jams, jellies and chutneys, try freezing or canning homemade fruit juices. Packets of frozen fruit are handy for baking or in smoothies, and dried fruit can be quickly rehydrated in warm water or munched as a snack.

Know Your Cold-Hardiness Zone

The "Zones" referred to in this article come from maps published by the United States Department of Agriculture that show the average minimum winter temperature for each region. Some types of fruit can tolerate more winter cold than others, so your area's cold-hardiness is important to know before you choose which fruit trees to grow. If you don't know your Zone, you can find it at the United States Department of Agriculture. If a mail-order nursery doesn't tell you which Zone a crop is suited for, you should probably buy from another supplier. To locate sources for fruit trees, check out our Seed and Plant Finder, which lets you quickly search the online catalogs of more than 500 mail-order nurseries and seed companies.

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